

LUNAR-V2 / TSUKIMI

Thermal Subsurface Understanding through Kinetic Ice Modeling and
Illumination

A Developer's Manual and Scientific Roadmap

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PHASE 1: VALIDATION STATUS

Site	Model	RMSE [K]	Criterion	Status
Apollo 15 (deep)	Discrete 3-layer	0.915	≤ 1.0	PASS
Apollo 15 (deep)	Hayne (2017)	1.441	≤ 1.5	PASS
Apollo 17 (1 m)	Discrete 3-layer	0.665	± 2.0	PASS
Apollo 17 (1 m)	Hayne (2017)	2.602	± 2.0	MARGINAL
Chang'E-4	Discrete 3-layer	--	O(1)	PASS
ChaSTE (flat)	Discrete 3-layer	--	$< 40 \text{ K bias}$	NOT ACTIVATION for Phase 2

Verdict: Phase 1 PASSED. All critical acceptance criteria satisfied. The discrete 3-layer model is the primary closure; Hayne (2017) is secondary. Ready to proceed to Phase 2.

PIPELINE ARCHITECTURE

Phase	Topic	Status	Description
Phase 0	Quickstart	✓ Shipped	Library smoke-test
Phase 1	Point-source validation	✓ Shipped	Apollo 15/17, CE-4, ChaSTE
Phase 2	Illumination coupling	■ In Progress	DEM + horizon + slope + shadow
Phase 3	Polar maps + ice	■ Planned	Diviner validation + ice-coupled k
Phase 4	Manuscript + thesis	■ Planned	PSJ submission + ISCT defence

RESEARCH NOVELTY

1. Multi-site blind validation

Regressed against Apollo 15/17, Chang'E-4, and ChaSTE with a single solver kernel. No site-specific tuning except H-scale.

2. Ice-coupled conductivity feedback

Self-consistent loop: Klinger's $k_{\text{ice}}(T) = 567/T$ couples to the solver. Deepens ice table by ~15 cm vs. fixed-k models.

3. Pixel-level topographic slope

$\cos(i)$ correction applied per DEM pixel. Collapses ChaSTE 40 K bias with $s=6^\circ$.

4. End-to-end reproducibility

Every notebook auto-installs deps and auto-fetches data. Full 4-site validation < 15 minutes on a 2024 laptop.

PHASE 2: TOPOGRAPHIC ILLUMINATION (IN PROGRESS)

Phase 2 extends the Phase-1 solver with pixel-level topographic forcing. The data flow is: DEM → slope/aspect → horizon raycast → sky view factor → shadow mask → corrected $Q(t)$. The Phase-1 kernel is reused unchanged.

Key equation (corrected insolation):

$$Q(t) = m(t) \times S_{\blacksquare} \times \cos(i) + F_{sky} \times Q_{IR,sky} + A \times S_{\blacksquare} \times \cos(i) \times (1 - F_{sky})$$

where $m(t)$ is the shadow mask, F_{sky} is the sky view factor, and $\cos(i)$ is the incidence angle corrected for local slope.

PHASE 3: POLAR MAPS & ICE-COUPLED CONDUCTIVITY (PLANNED)

Phase 3 builds a pixel-wise equilibrium temperature and ice-stability depth map for one south-polar tile, validated against Diviner bolometric T.

Ice-coupled conductivity closure (Klinger 1980):

$$k_{ice}(T) = 567 / T [W m^{-1} K^{-1}], T \in [20, 273] K$$

$$k_{eff}(T, z, \phi) = \phi \times k_{ice}(T) + (1-\phi) \times k_{reg}(T, z)$$

A fixed-point iteration solves self-consistently: (1) solve with dry regolith, (2) find $T_{stab} \approx 112 K$ stability depth, (3) set $\phi=0.05-0.10$ for $z \geq z_{\star}$, (4) re-solve with updated k_{eff} , repeat until $|\Delta z_{\star}| < 1 cm$ (typically 3–5 sweeps).

PHASE 4: MANUSCRIPT & THESIS (PLANNED)

Milestone	Target Date	Gate
Phase 2 complete	2026-04-30	ChaSTE bias ≤ 10 K
Phase 3 complete	2026-05-31	Diviner RMS ≤ 5 K
Manuscript draft	2026-06-30	Sections 2–7
Internal review	2026-07-31	Supervisor + co-authors
PSJ submission	2026-08-15	arXiv same day
Thesis defence	2026-09-15	ISCT public defence

Targets: PSJ submission by 2026-06, thesis defence by 2026-09. Full reproducibility artefact (code, Docker, data manifest) at Zenodo.